What Text Helps Forecast Inflation? Evidence from Retrieval-Augmented Language Models

Runxi Niu* Yuan Lin†

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Abstract

We examine which textual information improves forecasts of U.S. CPI inflation using a large language model (LLM). To limit data-leakage risk, the evaluation period begins after the model's documented pretraining cutoff. Our retrieval-augmented generation (RAG) framework provides the model with new information without retraining. It retrieves passages from central bank communications and financial news. In out-of-sample tests, the best RAG configurations outperform univariate benchmarks mainly at six and twelve months. Central bank communications are more helpful at one month, while news contributes more at twelve months. Several text-augmented prompts yield better-calibrated one-month prediction intervals than the raw numeric-only prompt. Topic-level analysis associates demand and energy content with larger forecast gains in some comparisons. Monetary policy content is generally associated with smaller gains.

Keywords: Inflation forecasting, large language models, retrieval-augmented generation, central bank communications, text as data, density forecasts.

JEL Classification: C53, E31, E37, E58.

*Department of Economics, Norwegian School of Economics (NHH), Bergen, Norway. Email: runxi.niu@nhh.no.

†School of Economics, Innovation, and Technology, Kristiania University of Applied Sciences, Bergen, Norway. Email: yuan.lin@kristiania.no.

1 Introduction

Inflation forecasting is central to economic decision-making, yet inflation has long been difficult to predict (Stock and Watson, 2007). A growing body of research suggests that textual data contains signals regarding future price changes—information that is difficult to encapsulate within a limited set of numerical indicators (Bybee et al., 2024; Larsen et al., 2021).

A key question is which types of text are useful for forecasting, and at which forecast horizons. Communications from central banks reflect policymakers' real-time assessments of economic conditions, while financial news captures the perspectives of journalists and market participants regarding shifts in supply, demand, energy, and labor markets.

We investigate this issue using large language models (LLMs), which are capable of processing economic text directly while accounting for word order and context. Traditional text analysis methods typically represent documents using pre-estimated topics or sentiment lexicons (Bybee et al., 2024; Loughran and McDonald, 2011), whereas LLMs can use text passages directly as inputs for forecasting. Recent applications include predicting financial returns (López-Lira and Tang, 2023) and forecasting inflation (Faria-e-Castro and Leibovici, 2024). However, once a model has been trained, the information embedded within its parameters does not automatically update in response to changing economic conditions.

Retrieval-Augmented Generation (RAG) provides models with documents from an external corpus that are available at each forecast origin. This allows newly released policy statements, economic reports, and news articles to be incorporated as model inputs without the need for retraining. Prompt templates dictate how forecasting instructions, numerical data, and retrieved text are presented. When comparing text sources, we keep the prompt format, numerical inputs, and forecasting instructions fixed. We vary only the source of the retrieved text.

We investigate how various types of text assist Large Language Models (LLMs) in forecasting inflation, and whether this utility varies according to the forecast horizon. At each forecast origin, we retrieve relevant passages from central bank communications and news articles, then feed these—along with historical Consumer Price Index (CPI) data—into the model. Through this design, we analyze how the choice of text source, the volume of retrieved text, and the presentation of numerical data influence forecasting accuracy.

RAG yields the largest improvements at six and twelve months, with more limited gains at shorter horizons. For each forecast horizon, we compare the best-performing RAG configuration with the best-performing univariate benchmark among those reported in Table 2. Each is selected by its RMSE on the evaluation sample.

For one-month forecasts, RAG reduces the RMSE by 2.9% and achieves a lower

squared error in 12 out of 25 forecast origins. However, excluding the forecast origin with the largest reduction in squared error reverses the RMSE ranking. For three-month forecasts, RAG achieves a lower squared error in 9 out of 23 forecast origins, yet its overall RMSE remains higher than that of an autoregressive benchmark model, AR(q).

For six-month forecasts, the RMSE decreases by 24%, with 14 out of 20 forecast origins showing a reduction in squared error. To determine whether this result is driven primarily by a few cases with substantial improvements, we excluded the three forecast origins with the largest gains; the advantage of RAG persists. For twelve-month forecasts, RAG achieves a 46% reduction in RMSE over the full evaluation sample compared to a three-month moving average benchmark. Reductions in error are also quite common: squared errors are lower at 10 of the 14 forecast origins.

We also compare RAG forecasts with those from the Survey of Professional Forecasters (SPF). To ensure a consistent basis for comparison, we calculate quarterly averages for both the forecasts and the actual inflation rates. Across all horizons, the best-performing RAG configurations achieve a lower root mean squared error (RMSE) than the SPF.

Text alone is not enough to produce accurate inflation forecasts. When we provide the LLM with only text, forecast errors are much larger than when we also include historical CPI data. How we present these numerical data also matters. A short statistical summary of CPI history can sometimes produce more accurate forecasts than a raw table. Together, these results suggest that text is more useful as a complement to historical inflation data than as a substitute for it.

The usefulness of text also depends on where it comes from and what it discusses. Central bank communications tend to provide more useful information at shorter forecast horizons, while news articles become more valuable at longer horizons. We also find that some topics are more closely associated with forecast improvements than others. In particular, when a larger share of the retrieved text discusses demand-related developments, forecast gains tend to be greater at the three-month and six-month horizons. This pattern appears in both central bank communications and news articles.

Not all types of text appear to be equally useful for forecasting inflation. We find that forecast gains tend to be smaller when a larger share of the retrieved text discusses monetary policy. One possible explanation is that LLMs have difficulty distinguishing between policy actions taken in response to inflation and information about future inflation. As a result, they may interpret policy responses as signals about where inflation is heading.

In addition to point forecasts, we examine the impact of text on the uncertainty surrounding inflation forecasts. At the one-month horizon, several text-augmented prompts produce lower Continuous Ranked Probability Score (CRPS) values and interval coverage closer to nominal levels than the raw numeric-only prompt. A more detailed numeric-only prompt also improves both measures.

Existing research has explored the roles of numerical data and text in macroeconomic forecasting. Bok et al. (2018) provide an overview of methods for nowcasting using large-scale numerical datasets, detailing how these approaches incorporate newly released data. Bybee et al. (2024) find that topics extracted from newspaper articles reflect macroeconomic fluctuations and contain information about future economic activity. In this paper, we feed retrieved text segments directly into a large language model (LLM) to investigate their influence on inflation forecasts, bypassing the need to first convert news articles into topic-based indicators.

Recent studies have also begun to explore the capacity of LLMs to extract useful economic information from text. López-Lira and Tang (2023) demonstrate that LLMs can identify sentiment in news headlines that correlates with stock returns. More closely related to our work, Faria-e-Castro and Leibovici (2024) provide evidence that LLMs can be used to forecast inflation. Building on this, we further investigate how the predictive value of retrieved information varies according to the source of the text, its subject matter, and the forecast horizon.

Bybee (2023) uses news articles to prompt LLMs to generate economic expectations, subsequently comparing these outputs against survey data. Our study incorporates document retrieval and filtering at each forecast origin, while evaluating both point and density forecasts. By adjusting numerical inputs and retrieval parameters, we observe how the LLM's forecasts change. Comparing different sources and topics helps identify which types of textual content are associated with improvements in forecast performance.

Another strand of research focuses on whether language model architectures can use numerical time series to forecast economic variables. Carriero et al. (2025) compare models pre-trained on numerical sequences against traditional forecasting methods, such as Bayesian vector autoregressions (BVARs) and factor models. The best-performing pre-trained models achieve a level of predictive accuracy comparable to these benchmarks. However, the prediction accuracy of all models declined between 2020 and 2022. This paper examines how LLMs use historical inflation data combined with retrieved text to generate forecasts. In our post-pandemic sample from January 2024 to February 2026, the best-performing RAG configurations outperform the selected univariate benchmarks at six and twelve months.

Data leakage occurs when an LLM's training data include the inflation outcomes used to evaluate its forecasts. Its apparent forecasting accuracy may then reflect memorization. Faria-e-Castro and Leibovici (2024) evaluate Google PaLM on inflation forecasts over a period that largely overlaps with the model's training data. Alam et al. (2026) show that ChatGPT's pseudo-out-of-sample forecasts appear competitive, but its true out-of-sample predictions deteriorate sharply. To reduce the risk of data leakage, we use the open-weights Llama-3-8B-Instruct model, whose documented pretraining cutoff is March 2023. We also probe its knowledge of historical CPI releases as a supplementary

check. Our evaluation begins in January 2024, after the documented pretraining cutoff.

The rest of the paper is organized as follows. Section 2 describes the data and forecasting framework. Section 3 evaluates forecast accuracy across different horizons. Sections 4 and 5 examine the value of textual information, first by comparing text sources and topics, and then by comparing RAG with a topic-model forecasting benchmark. Sections 6 and 7 investigate whether text provides early signals of inflation and whether it improves density forecasts and uncertainty calibration. Section 8 concludes.

2 Data and Methodology

2.1 Forecast Design

Our target variable is the year-over-year percentage change in the U.S. Consumer Price Index for All Urban Consumers (CPI-U), observed at monthly frequency. We forecast inflation for horizons of $h \in \{1, 3, 6, 12\}$ months. The CPI dataset begins in January 1990. At each forecast origin, we use only the observations that have been released up to that date.

We evaluate the LLM in a zero-shot setting, without re-estimating its parameters. For the numerical benchmark models that require estimation, we refit the models at each forecast origin using a rolling 24-month window of data available at that time. The forecast origin is defined as the official release date of the U.S. Bureau of Labor Statistics (BLS) CPI report for the current month—the date on which the previous month's CPI data becomes publicly available.[1] See Section 2.3 for details on date constraints during the data retrieval process.

The evaluation period spans forecast origins from January 2024 to February 2026. For the one-month horizon, there are up to 25 evaluable monthly origins. For longer horizons, fewer origins are evaluable because some actual values are not yet available: 23 origins for the three-month horizon, 20 for the six-month horizon, and 14 for the twelve-month horizon.[2] The evaluation period begins after the model's documented pretraining cutoff (see Section 2.6).

Details regarding the estimation and hyperparameter tuning methods for each model are provided in Appendix C. For Autoregressive (AR) and Autoregressive Integrated Moving Average (ARIMA) models, the Akaike Information Criterion (AIC) is used for order selection, based solely on observations within the current estimation window. Specifications that use cross-validation also have their hyperparameters tuned within this same

[1] We use the BLS publication schedule (`https://www.bls.gov/schedule/news_release/cpi.htm`) to pin each origin to the actual release day, which varies across months (typically between the 10th and 15th).

[2] The Bureau of Labor Statistics (BLS) did not publish CPI data for October 2025 because of the lapse in federal funding. We therefore exclude forecasts for October 2025 from the accuracy evaluation.

window. Note that a 24-month window does not necessarily yield 24 observations available for estimation; the sample size may be reduced by missing values, and some observations are lost when constructing lagged inputs.[3] RMSE is computed as $\sqrt{N^{-1}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2}$, where N is the number of scored forecasts for the model and horizon.

2.2 Text Corpus

We use two sources of text: central bank communications (CB) and news articles, covering January 2023 to March 2026.

The central bank collection contains 2,172 documents from official sources. These include speeches, testimony, and press releases from the Federal Reserve; meeting minutes and statements from the Federal Open Market Committee (FOMC); and speeches from the European Central Bank (ECB) and the Bank of England (BOE). We also include central bank speeches collected by the Bank for International Settlements (BIS) and articles from the International Monetary Fund (IMF).[4]

The second pool, financial news (News), consists of 10,225 financial news articles drawn from a global event database.[5] News reflects market participants' expectations. It also covers changes in commodity prices, labor markets, and fiscal policy. These changes may appear in news reports before they are reflected in official data.

We standardized the spacing and date formats across both document pools, allowing us to apply a consistent publication cutoff across all sources at each forecast origin. We also removed duplicate documents.[6] News articles underwent additional quality filtering to eliminate entries with low content or redundancy. Specifically, we discard news articles if alphabetic characters account for less than 55% of all characters, if the combined share of the delimiters `#`, `;`, and `|` exceeds 2%, or if their source domain is classified as low quality. These rules help filter out text of poor extraction quality, as well as records

[3]We divide forecast origins into three periods: training (before January 2022), validation (January 2022 to December 2023), and evaluation (January 2024 onward). These periods are used to organize and report the results. They do not determine which observations are used to estimate each forecast. For the elastic net and random forest models that use cross-validation, we use time-based folds within the estimation window available at each forecast origin. Our primary forecast comparison is based on the root mean squared error (RMSE) calculated over the evaluation set. The LLM used is a frozen, pretrained model; it is not trained on these data.

[4]Documents from the Federal Reserve and the Bank for International Settlements (BIS) were collected via Really Simple Syndication (RSS) feeds and annual ZIP archives; European Central Bank (ECB) speeches were obtained through a comma-separated values (CSV) feed; Bank of England speeches were retrieved via RSS; and International Monetary Fund (IMF) articles were acquired through manual export.

[5]We use the GDELT (Global Database of Events, Language, and Tone) project, querying the GDELT Doc application programming interface (API) and 15-minute snapshots from the Global Knowledge Graph. Articles were selected using the query `inflation OR ‘‘consumer prices’’ OR CPI OR deflation`, restricted to U.S.-located coverage, and matched against the theme keyword `INFLATION`. Full-text content was extracted from the available source URLs.

[6]Deduplication was performed using stable document identifiers, constructed by applying the Secure Hash Algorithm 1 (SHA-1) to source metadata (including source type, name, title, publication date, URL, and text).

containing very little readable content. We identify duplicate news reports using URL paths or, when unavailable, hashes of normalized titles and text.[7] Only articles that pass these screening criteria are included in the retrieval corpus.

Furthermore, documents are segmented into overlapping chunks of 1,200 characters, with an overlap of 200 characters; the segmentation points are aligned with word boundaries.[8] This approach enables the retrieval process to extract relevant passages from long documents, while the overlapping segments preserve contextual information at the boundaries between chunks.

The final corpus available for retrieval consists of 70,840 chunks: 37,536 from central bank sources and 33,304 from news sources. Publication dates are stored at the chunk level and enforced during the retrieval process (Section 2.3).

2.3 RAG Workflow

For each forecast origin date t and forecast horizon h, the retrieval-augmented generation (RAG) workflow constructs a prompt in four stages, as illustrated in Figure 1.

In the first stage, we construct a natural-language query specifying the target month and the year-over-year headline CPI inflation rate, using TF-IDF to represent both the query and the document chunks. Appendix L provides an example of a complete retrieval query for a representative forecast origin.[9]

In the second stage, we exclude document chunks whose parent documents are dated after the forecast origin, and by default, we apply a 60-day recency window. We then retrieve the top k eligible document chunks based on text similarity (with $k = 20$ by default).[10]

In the third stage, we re-rank the retrieved text chunks to select those most relevant to the forecast. The ranking considers more than just how closely a chunk matches the query. It also takes into account when the text was published, what topic it discusses, and, for news articles, several measures of quality.

More recent texts receive higher weights, with the recency weight declining exponen-

[7] The canonical story identifier is derived from the web address path or, when unavailable, from a normalized hash of the article title and text.

[8] Each chunk receives a stable identifier (SHA-1 hash of the parent document identifier, chunk index, and chunk text) and inherits its parent document's metadata: publication date, source name, institution, country, and document type.

[9] TF-IDF stands for term frequency–inverse document frequency. TF-IDF assigns weights to terms in the query and each passage. Terms that appear in many documents receive less weight when measuring similarity. The vectorizer uses a vocabulary of 50,000 terms, unigram and bigram features, English stop-word removal, and a minimum document frequency of 2. Retrieval scores are cosine similarities between query and chunk TF-IDF vectors.

[10] The retrieval window (60 days) and chunk size (1200 characters with 200-character overlap) were set based on domain considerations rather than tuned on the evaluation sample. Appendix P uses an expanded pool of 48 candidates as the reference for examining retrieval-set stability under alternative candidate pool sizes, reranked set sizes, retrieval windows, and chunk sizes.

tially and halving every 30 days. We also adjust the scores of texts discussing particular topics, such as demand, energy, and monetary policy. For news articles, we further account for text completeness, the reliability of the source, and whether the story is unique.[11]

Formally, given a query q and a set of retrieved chunks $C = \{c_1, c_2, \ldots, c_k\}$, the weight assigned to each chunk c_i is determined by its relevance score r_i and reliability score ρ_i:

$$w_i = \frac{r_i \cdot \rho_i}{\sum_{j=1}^{k} r_j \cdot \rho_j}, \tag{1}$$

where r_i denotes the composite reranking score (base similarity $\times$ recency multiplier $\times$ topic-channel weight) and ρ_i captures source-specific quality adjustments (text-quality multiplier $\times$ domain-quality multiplier $\times$ duplicate-story penalty). After reranking, we retain the top-k' chunks (default $k' = 8$) and select three text extracts from them for inclusion in the LLM prompt.

In the final stage, text selected from the retrieved chunks is combined with historical CPI data and forecasting instructions. We call the text included in the prompt the retrieved context. A chunk is a document segment used in retrieval; text extracts are selected from these chunks for presentation to the LLM. We use text summary (or RAG summary) to refer to the retrieved context formed by these extracts. The resulting prompt is submitted to the model to generate a forecast.

2.4 Prompt Design

We vary numerical inputs and retrieved text to compare forecast accuracy. Retrieval uses CB, News, CB+News, or All. CB+News combines separate searches of the two pools; All searches them jointly. Table 1 maps out the core year-over-year prompt families along these two dimensions.[12]

Along the numeric dimension, we use five input forms: (a) None: no CPI history; (b) Stat summary: a short summary of recent CPI data; (c) Full table: the 24 most recent monthly CPI readings; (d) Stat + table: the statistical summary together with the full table; (e) Detailed + table: an expanded statistical summary together with the full table.

We vary whether the prompt includes retrieved text and, when it does, which passages are shown to the LLM. Each passage is accompanied by its source name and publication

[11] All settings are heuristic. Topic multipliers: 1.25 (energy, housing, labor, demand, supply) and 0.6 (monetary policy). Recency decays exponentially with a 30-day half-life (7 days for $h = 1$). News multipliers: 0.15 (metadata-only text), 0.7 (low-quality domains), 0.5 (duplicates), and 0.2 (noisy sources). Appendix H tests alternative settings.

[12] Additional variants targeting CPI index levels are documented in Appendix A. Results for prompt variants 1–14 are reported in Appendix B.

date. Appendix N reports how many text chunks are retrieved and how many extracts are included in the prompts.

These choices give us 33 prompt variants for forecasting year-over-year inflation. Variant 1 serves as our baseline and includes the full CPI table without any text. Variants 11–14 add text from different sources while keeping the CPI table unchanged.

To compare text sources more directly, we use groups of variants such as 19–21 and 25–27. Within each group, the prompt format remains the same, and only the source of retrieved text changes. This allows us to compare central bank communications, news articles, and their combination without changing the rest of the prompt. Appendix N provides examples of the prompts.

2.5 Numeric Benchmarks

To contextualize the LLM forecasts, we include benchmarks spanning naive forecasts, linear time-series models, multivariate macroeconomic models, and machine-learning approaches.

At the simplest end, we include three naïve benchmarks: the last-observed value (random walk in levels), the 3-period mean, and the seasonal naïve forecast (same month, prior year).

For linear time-series models, we estimate an autoregressive model AR(q) with lag order $q \in \{1, \ldots, 6\}$ chosen by AIC at each forecast origin. The AR(q) specification is:

$$\hat{y}_t = \phi_0 + \sum_{i=1}^{q} \phi_i \, y_{t-i} \,, \tag{2}$$

where $\phi_0, \phi_1, \ldots, \phi_q$ are the estimated model coefficients.

We also estimate an automatically selected ARIMA specification (order chosen by AIC). The general ARIMA(p, d, q) model is:

$$\Delta^d y_t = \phi_0 + \sum_{i=1}^{p} \phi_i \, \Delta^d y_{t-i} + \sum_{j=1}^{q} \theta_j \, \varepsilon_{t-j} + \varepsilon_t \,, \tag{3}$$

where Δ^d denotes the d-th difference operator and ε_t is the error term at time t.

A bivariate Phillips curve regresses inflation on the unemployment gap, defined as the deviation of the unemployment rate from the Congressional Budget Office (CBO) natural rate estimate, and one lag of inflation.

A vector autoregression (VAR) including inflation, the unemployment rate, and the federal funds rate, with lag order selected by AIC, accommodates multivariate dynamics.

The VAR model is expressed as:

$$\mathbf{y}_t = c + \sum_{i=1}^{p} A_i \, \mathbf{y}_{t-i} + \varepsilon_t \,, \tag{4}$$

where $\mathbf{y}_t$ is a vector of variables (inflation, unemployment rate, federal funds rate), A_i are the coefficient matrices, and ε_t represents the error terms at time t.

We also compare the LLM forecasts with two machine-learning models: an elastic net and a random forest (RF). The elastic net provides a regularized linear model, while the random forest allows for nonlinear relationships.

Both models use the three most recent lags of four variables: headline CPI inflation, core CPI inflation, the unemployment rate, and the federal funds rate. For longer forecast horizons, we generate forecasts recursively. We keep core inflation, unemployment, and the federal funds rate fixed at their latest observed values.

In our main results, we use fixed hyperparameters for both models. Appendix C describes the model specifications, while Appendix J compares the results with models whose hyperparameters are selected using time-based cross-validation.

We also include a quarterly proxy derived from the Philadelphia Fed's Survey of Professional Forecasters (SPF). The SPF mean CPI-level path is converted to year-over-year inflation and mapped to monthly forecast origins using the most recent survey available at each origin date. SPF forecasts are quarterly. We assign the same forecast to all three months of the target quarter. For twelve-month forecasts, we adapt the benchmark of Atkeson and Ohanian (2001). We average the last 12 monthly observations of year-over-year inflation. Appendix C explains how our version differs from the original.

2.6 Large Language Model

Throughout the paper we use Meta's Llama-3-8B-Instruct. This model was released on April 18, 2024 and was pretrained on over 15 trillion tokens of publicly available data with an official knowledge cutoff of March 2023.[13] We choose this open-weights model because its documented pretraining cutoff helps us assess the risk that forecasts reflect memorized outcomes (Section 1). We also probe the model's knowledge of historical CPI releases by asking about specific CPI observations and its most recent available data (Appendix K).

We evaluate the forecasts from January 2024 onward, after the model's documented pretraining cutoff.

For point forecasts, we set the temperature to zero. This means that the LLM selects the most likely next token at each step, rather than randomly sampling from possible

[13]Source: Meta's model card at `https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct`. The instruct variant was further fine-tuned on over 10 million human-annotated examples; no exact post-training completion date is provided.

outputs. Using this setting reduces randomness in the forecasts and makes it easier to compare different prompt designs and text sources. For density forecasts, we ask the LLM to report predictive quantiles in a single response.

2.7 Evaluation Metrics

For point forecasts, the main tables report root mean squared error (RMSE) over the evaluation sample. Appendix B reports RMSE, mean absolute error (MAE), and mean forecast error (ME; bias) for prompt variants 1–14 at all four horizons.

We evaluate density forecasts using several measures of accuracy and uncertainty calibration. The Continuous Ranked Probability Score (CRPS; Gneiting and Raftery, 2007) and the log predictive score (Mitchell and Hall, 2005) are used to measure how well predictive distributions align with actual outcomes.

Interval coverage is used to assess the calibration of prediction intervals. We calculate the proportion of actual values that fall within prediction intervals with nominal coverage levels of 68% and 90%, while also reporting the average width of these intervals.

Only forecasts for which the actual values are known are included in the evaluation. The tables report the sample size and indicate which comparisons are based on the same forecast origins.

Table 1: Overview of Prompt Variants

	Text input	
Numeric input	None	Retrieved text
None	—	3^{All}, 16–$18^{\text{CB/N/B}}$
Stat summary	2	4^{All}, 15^{N}, 19–$21^{\text{CB/N/B}}$
Full table	1	6^{All}, 22–$24^{\text{CB/N/B}}$, 11^{All}, 12^{CB}, 13^{N}, 14^{B}
Stat + table	5	7^{All}, 25–$27^{\text{CB/N/B}}$
Detailed + table	8	9–10^{All}, 28–$33^{\text{CB/N/B}}$

Notes: Each cell shows the corresponding prompt variant IDs. Superscripts indicate which text sources are allowed: All = all sources; CB = central bank communications only; N = news only; B = central bank communications and news. All searches the full corpus jointly; B combines separate searches of central bank communications and news. Source labels describe the allowed retrieval pools, rather than the source composition of the retrieved text in a particular run. Variant 15 uses a filtered news summary with statistical inputs at $h = 1$ only. Additional horizon-specific filtered variants (34–37) and alternative-target variants (year-over-year inflation (YoY), CPI level) are listed in Appendix A.

3 Does Retrieval-Augmented Text Improve Inflation Forecasts?

We examine whether LLM forecasts with retrieved text outperform numerical benchmarks.

3.1 Comparison of RAG-Augmented Large Language Models (LLMs) and Numerical Benchmark Models

Table 2 presents the root mean squared error (RMSE) for various LLM configurations and numerical benchmark models across different forecast horizons ($h \in \{1, 3, 6, 12\}$ months). The results indicate that model performance improvements are most pronounced at the 6-month and 12-month horizons, whereas improvements at shorter horizons are smaller and less consistent.

For shorter forecast horizons, the best-performing RAG-augmented configuration in Table 2 (variant 4: incorporating both statistical and textual summaries) yields an RMSE of 0.174 at $h = 1$, slightly outperforming the AR(q) benchmark (0.179) and the ARIMA model (0.188). This configuration also outperforms the LLM prompting approach that relies solely on numerical inputs (variant 1, with an RMSE of 0.220), though this comparison involves variations in both the presentation of numerical data and the inclusion of textual input. The marginal advantage over the AR(q) model aligns with findings from previous research, which suggest that recent data holds predictive value for inflation one month ahead (Stock and Watson, 2007). At $h = 3$, the best-performing LLM configuration in Table 2 is the statistical summary without retrieval (variant 2, RMSE = 0.371), which performs roughly on par with the ARIMA model (0.372). Interestingly, incorporating retrieved text into this summary does not improve accuracy; variant 4 yields an RMSE of 0.417.

At longer forecast horizons, the advantage becomes substantial. For $h = 6$, variant 4 achieves an RMSE of 0.268, compared to 0.352 for the AR(q) model and 0.466 for the ARIMA model—representing a 24% reduction in RMSE relative to the best-performing numerical benchmark model. For $h = 12$, the RAG configuration that retrieves only news text (variant 13) yields an RMSE of 0.284, compared to 0.524 for the three-month mean benchmark—a reduction in RMSE of 46%. At this forecast horizon, all RAG configurations in Table 2 that incorporate numerical inputs outperform the numerical benchmarks.

Figure 2 illustrates the squared prediction errors for the best-performing RAG configuration (from those listed in Table 2) and the best-performing univariate benchmark model for each forecast horizon $h \in \{1, 3, 6, 12\}$. We select models separately at each horizon using each candidate's available evaluation-sample RMSE and hold the selected models

fixed across all plotted forecast origins. Based on this criterion, variant 4 is selected for horizons $h = 1$ and $h = 6$, while the news-only retrieval specification—designated as variant 13—is chosen for horizons $h = 3$ and $h = 12$. The selected benchmark model is an AR(q) model for $h \in \{1, 3, 6\}$, and the three-month mean benchmark for $h = 12$. Table 18 specifically examines the performance of variant 4 across all four forecast horizons.

For $h = 1$, the RAG model yields a lower squared error in 12 out of the 25 forecast origins, resulting in an overall 2.9% reduction in RMSE (from 0.179 to 0.174). However, if the forecast origin of March 2024—which accounts for the largest improvement—is excluded, the ranking reverses: the RAG model's RMSE becomes 0.170, compared to 0.153 for the AR(q) model. For $h = 3$, the RAG model produces a lower squared error in 9 out of the 23 forecast origins, yet its overall RMSE is higher than that of the AR(q) model (0.401 versus 0.366). Notably, the forecast origin of June 2024 (targeting August 2024) accounts for approximately 89% of RAG's total net excess squared error for this horizon. Taken together, these results suggest that the selected RAG model offers only limited gains for short-term forecast horizons.

For $h = 6$, the RAG model yields a lower squared error at 14 out of the 20 forecast origins, with an overall 24% reduction in RMSE (from 0.352 to 0.268). Excluding the three forecast origins that show the greatest improvement, the RMSE becomes 0.246 (compared to 0.282 for the AR(q) model). Thus, at $h = 6$, the benefits of RAG are not only observed across the majority of forecast origins but also persist even after removing the cases with the largest gains.

For a forecast horizon of $h = 12$, RAG achieves a lower squared error at 10 out of the 14 forecast origins; compared to the three-month mean benchmark, RMSE is reduced by 46% (0.284 versus 0.524). Approximately 98% of the total net reduction in squared error is attributable to the forecast origins from April to June 2024 (for forecasts covering March to May 2025). Excluding these three forecast origins, the RMSE becomes 0.310 (compared to 0.318 for the benchmark). On average, the improvements for the twelve-month forecast horizon are substantial and evident across most forecast origins, though the bulk of these gains is concentrated in those three specific instances. All four subplots present the complete set of pairwise evaluation observations.

Appendix B.1 reports the performance of the same model pairs using mean absolute error (MAE) and mean forecast error metrics. Under the MAE metric, improvements for the six-month and twelve-month horizons remain, with error reductions of 29.4% and 44.8%, respectively. For the one-month horizon, the ranking reverses under the MAE metric, while for the three-month horizon, RAG's forecast accuracy remains lower than that of the AR(q) model.

We also compare the predictions from LLMs with those from the Federal Reserve Bank of Philadelphia's Survey of Professional Forecasters (SPF). To align with the SPF's reporting frequency, we aggregate all forecast data into a quarterly format and evaluate

RMSE accordingly. Full results are provided in Appendix M. The best-performing LLM variant outperforms the SPF across all forecast horizons, with the most substantial improvement observed at $h = 6$ (a 64% reduction in RMSE), followed by $h = 12$ (a 49% reduction). Depending on the forecast horizon, these comparisons encompass between 3 and 7 non-overlapping quarters.

3.2 Ablation Study: Where Do the LLM's Advantages Come From?

To identify the source of these forecasting improvements, we compare several approaches: raw numerical inputs, structured summaries, and retrieval-augmented prompts.

We first examine the role of numeric grounding. The baseline LLM prompt (variant 1), which provides raw CPI tables directly, without retrieval or structured summarization, yields an RMSE of 0.220 for a forecast horizon of $h = 1$, which is higher than the 0.179 RMSE of the AR(q) benchmark model. For longer horizons of $h = 6$ and $h = 12$, the RMSE rises to 0.470 and 0.518, respectively—levels comparable to those of simple, naïve forecasting benchmarks. This indicates that prompts providing only raw tables offer no consistent accuracy advantage over standard numerical benchmark models.

In contrast, replacing the raw tables with a concise statistical summary—containing the most recent figures and three-period averages (variant 2)—leads to substantial gains in long-horizon forecasting. At $h = 1$, the RMSE drops to 0.218 (a marginal improvement), while at $h = 6$ and $h = 12$, the RMSE decreases to 0.275 and 0.334, respectively. For $h = 12$, the RMSE falls from 0.518 to 0.334—a 36% reduction—showing that numerical presentation matters for forecast accuracy. Structured summaries may help the model identify relevant features of the inflation process without retrieving external text.

Forecasting performance can be further enhanced by supplementing statistical summaries with text summaries obtained through a retrieval process (variant 4: statistical summary + RAG-based text summary). In this case, the RMSE values for $h = 1$, $h = 6$, and $h = 12$ improve to 0.174, 0.268, and 0.318, respectively. For the 6-month and 12-month forecast horizons, the incremental gain from retrieved text is smaller than the improvement achieved by replacing raw tables with statistical summaries. Specifically, at $h = 6$, transitioning from statistical summary only to statistical summary + text summary reduces the RMSE by approximately 3%, whereas the transition from raw tables to statistical summary yields a 42% reduction in RMSE.

As a diagnostic test, we also examine whether an LLM can generate forecasts relying solely on retrieved text—without any historical CPI data (variant 3). In this setup, the RMSE is 1.984 for $h = 1$ and 1.961 for $h = 3$; by comparison, the corresponding values for an AR(q) model are 0.179 and 0.366, respectively. At $h = 12$, the RMSE using only text-based inputs is 1.309—a figure higher than those obtained by the numerical benchmark

models or the various LLM configurations that incorporate numerical inputs (as shown in Table 2). In these experiments, prompts containing only text yield much larger errors than those that also include historical CPI data.

Table 18 presents a comparison of prediction accuracy, along with results from the Diebold–Mariano (DM) test (Diebold and Mariano, 1995). In Panel A, RAG variant 4 is held constant (remaining the same across all forecast horizons) and compared against the univariate benchmark model with the lowest available evaluation-sample RMSE at each horizon: AR(q) at $h \in \{1, 3, 6\}$ and the three-month mean benchmark at $h = 12$. The two-sided DM tests use squared-error loss with Newey–West heteroskedasticity and autocorrelation consistent (HAC) variance. At $h = 1$, variant 4 has a slightly lower RMSE than AR(q), but the difference is insignificant ($p = 0.829$). At $h = 3$, its RMSE is higher than AR(q), although this difference is also insignificant ($p = 0.180$). These short-horizon results are consistent with the difficulty of beating simple autoregressions (Stock and Watson, 2007). At $h = 6$ and $h = 12$, the lower squared-error loss of variant 4 is significant at the 5% level ($p = 0.028$ and $p = 0.024$, respectively).

Panel B compares the raw numeric-only prompt (variant 1) with a prompt combining statistical summaries and retrieved text (variant 4). This comparison changes both numerical presentation and text input. Variant 4 has lower RMSE at $h = 1, 6, 12$ and higher RMSE at $h = 3$.

Panel C compares the forecast accuracy of CB-only and news-only statistical-summary prompts (variants 19 and 20). None of the four horizons rejects equal predictive accuracy at conventional significance levels. The direction of the DM statistic shifts from favoring CB at short horizons to favoring news at $h = 12$ ($p = 0.137$), consistent with the source crossover pattern documented in Section 4. The comparisons cover 14–25 forecast origins: the source rankings vary across horizons, but the differences are not statistically significant in this sample.

3.3 Comparison of Prompt Designs

We compare prompts that combine retrieved text with different presentations of CPI history.

The statistical-summary prompt (variant 4) has lower RMSE than the table prompt (variant 11) at every horizon. The prompt with a detailed summary and table (variant 9) also beats the table prompt, except at three months.

A summary may help the LLM identify useful information in CPI history. However, the retrieved passages also differ across prompts. We are cautious about attributing these gains to numerical format or text length alone.

Table 2: Main Forecasting Results: RAG-Augmented LLMs vs. Numeric Benchmarks (Evaluation Sample)

		Forecast Horizon (Months)			
	Variant	$h = 1$	$h = 3$	$h = 6$	$h = 12$
Panel A: LLM Variants					
LLM	4: Stat summary + RAG summary (CB)	**0.174**	0.417	**0.268**	0.318
LLM	8: Detailed preamble + table	0.214	0.424	0.278	0.299
LLM	7: Stat + RAG summary + table (All)	0.316	0.471	0.303	0.312
LLM	9: RAG summary + detailed table (All)	0.248	0.513	0.339	0.300
LLM	2: Stat summary only	0.218	**0.371**	0.275	0.334
LLM	1: Numeric table only	0.220	0.388	0.470	0.518
LLM	11: RAG text extracts (All)	0.293	0.465	0.457	0.468
LLM	12: RAG text extracts (CB)	0.266	0.487	0.475	0.350
LLM	13: RAG text extracts (News)	0.276	0.401	0.389	**0.284**
LLM	14: RAG text extracts (CB+News)	0.276	0.500	0.494	0.334
LLM	3: RAG summary only (CB)	1.984	1.961	0.674	1.309
Panel B: Numeric Benchmarks					
Benchmark	AR(q)	0.179	0.366	0.352	0.662
Benchmark	ARIMA (auto)	0.188	0.372	0.466	0.557
Benchmark	Phillips Curve	0.220	0.412	0.508	0.852
Benchmark	Elastic Net	0.255	0.463	0.603	0.596
Benchmark	Random Forest	0.266	0.400	0.486	0.530
Benchmark	VAR (AIC)	0.326	0.679	1.027	1.288
Benchmark	Naïve (last obs.)	0.195	0.395	0.434	0.541
Benchmark	Naïve (3-period mean)	0.274	0.402	0.403	0.524
Benchmark	Seasonal Naïve	1.120	0.753	0.620	0.541

Notes: All values are RMSE on the evaluation sample. All LLM variants use Llama-3-8B-Instruct. In Panel A, bold values indicate the lowest RMSE achieved by any LLM variant for each forecast horizon. Variant numbering is consistent with Table 1. Regarding prompt design, Variants 3 and 4 allow for the use of all available text sources; however, the forecasts reported here are based specifically on central bank texts.

Variants 4, 7, 9, and 11–14 incorporate both numerical inputs and text summaries, with the latter generated from retrieved text extracts. Variants 1, 2, and 8 rely exclusively on numerical inputs, while Variant 3 uses only retrieved text. Comparisons with the Survey of Professional Forecasters (SPF) are conducted on a quarterly basis; see also Appendix M. Evaluation window: January 2024–February 2026. At $h = 1, 3, 6, 12$, evaluated forecast origins start in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively ($N = 25, 23, 20, 14$). Only target outcomes observed by February 2026 enter the evaluation. The target months corresponding to each horizon begin in January, March, June, and December 2024, respectively. For the last-observed and three-month mean benchmarks, one additional forecast is missing for both the one-month and three-month horizons, resulting in sample sizes of $N = 24, 22, 20, 14$.

4 Central Bank Communications vs. Financial News: A Source-Based Analysis

We employ a fixed-prompt design to compare central bank communications with financial news. We then investigate the relationship between the topics present in the retrieved text and improvements in predictive performance (i.e., forecast gains).

4.1 Comparison by Source

To compare the information provided by different text sources, we hold the numerical inputs and forecasting instructions constant while varying the source of the retrieved text. Table 4 presents the results for four prompt designs, using central bank communications, financial news, and a combination of both. The predictive value of a text source varies according to the forecast horizon. When summaries of recent CPI data are included in the prompt, central bank texts yield smaller errors than financial news for one-month-ahead forecasts. This advantage narrows at the three-month and six-month horizons, while for twelve-month forecasts, financial news performs better (Figure 3).

This reversal in the ranking of sources occurs across all four prompt designs. For the three-month and six-month horizons, the relative ranking of sources depends on the specific prompt design.

Central bank communications summarize current economic conditions and policy decisions, which may help an LLM interpret recent CPI data. Financial news, by contrast, covers shifts in spending, production, and costs—changes that may take longer to manifest in consumer prices.

4.2 Thematic Dimensions and Predictive Gains

We categorize the retrieved texts into six topics: demand, energy, labor, housing, supply, and monetary policy, based on keywords. Each retrieved chunk is assigned a primary topic, and we calculate topic shares at each forecast origin.[14]

Table 5 reports the correlation between topic shares and forecast improvements. At each forecast origin, forecast improvement is the absolute error of the numeric-only LLM baseline (variant 1) minus the RAG absolute error. A positive correlation indicates that a higher share of a given topic is associated with a greater reduction in forecast error.

Whether derived from central bank texts or financial news, content related to "demand" shows a positive correlation with forecast improvements in several comparisons. Reports on spending and retail activity may help the LLM identify demand-side pressures

[14]The classifier uses a manually curated list of keywords for each topic and assigns text chunks to the topic with the highest keyword density. Chunks that fail to meet a minimum keyword threshold are labeled as "Other" and are excluded from topic analysis.

not yet fully captured in recent CPI data; such text may provide supplementary context to the numerical data included in the prompt.

For six-month-ahead forecasts, content concerning "energy" in central bank texts is also associated with greater forecast improvements. One possible explanation is that discussions regarding production and transportation costs offer insights into price pressures before they propagate to consumer prices. However, the strength of this association varies depending on the text source and the forecast horizon.

The results for "monetary policy" differ. In most comparisons, a higher share of text discussing monetary policy is associated with smaller forecast improvements—with the exception of twelve-month forecasts using financial news. The utility of information regarding monetary policy appears to depend on both the source of the text and the forecast horizon.

4.3 Interpreting Monetary Policy-Related Content

Texts concerning monetary policy typically discuss both inflationary pressures and the measures taken by the central bank to address them. Since these two types of information may carry different implications for future inflation, they can be challenging to interpret. Romer and Romer (2004) draw a similar distinction: between "monetary policy shocks" and the systematic policy responses central banks implement based on their assessment of future economic conditions.

For instance, a "hawkish" central bank statement might highlight persistently high inflation while simultaneously outlining contractionary policies intended to curb it. An LLM might focus on the warnings about inflation, failing to fully account for the anticipated impact of the policy response. Consequently, the model could interpret the statement as a signal of higher future inflation, even though the policy measures themselves are designed to drive inflation down.

As monetary policy is a recurring theme in central bank communications, the way such content is handled affects which passages are retrieved. Our re-ranking approach allows for the down-weighting of specific chunks during the selection process (see Section 2.3). Building on the correlation results presented in Table 5, we further investigate whether topic weights can be adjusted through online learning.

4.4 Online Learning of Topic Weights

The correlation coefficients shown in Table 5 are calculated *ex post* using the full evaluation sample—data that would not be available at the time of any given forecast. To examine how topic weights evolve in real time, we use an online updating method, adjusting the weights only after the corresponding actual CPI figures are released.

At each forecast origin t, the retriever calculates the distribution of each retrieved chunk across the six defined topics. Once the actual value of the target CPI is released, the system updates the weight vector $\mathbf{w}_t \in \Delta^{K-1}$—where $K = 6$ and Δ^{K-1} denotes the probability simplex—according to the following equation:

$$\mathbf{w}_{t+1} = \Pi_\Delta(\mathbf{w}_t + \eta \cdot r_t \cdot (\mathbf{s}_t - \mathbf{w}_t)), \tag{5}$$

Here, $\mathbf{s}_t$ represents the average topic distribution of the text chunks retrieved at time t, weighted by their retrieval scores and normalized such that the components sum to 1.

The reward signal is determined by the squared prediction error:

$$r_t = \kappa - (f_t - y_t)^2.$$

We set a fixed threshold of $\kappa = 1.0$. The reward is positive when the squared error falls below this threshold, and negative when it exceeds it. The learning rate is set to $\eta = 0.25$. The operator Π_Δ denotes the Euclidean projection onto the probability simplex, ensuring that the updated weights remain non-negative and sum to 1.[15]

All topics share the same reward value, r_t. When the reward is positive, the weights shift toward the topic distribution of the retrieved chunks. While this update reflects which topics were present when accurate predictions occurred, it cannot distinguish which topics actually contributed to the improvement in prediction accuracy.

The initial weight for each channel k is $w_0^k = 1/K \approx 0.167$. The updated weights are then used to rank retrieved passages at subsequent forecast dates. The adjusted score is

$$\tilde{s}_i = s_i + \alpha \sum_k w_t^k \cdot d_i^k.$$

Here, d_i^k is the share of topic k in text segment i. We set $\alpha = 0.25$ to control the strength of the reranking adjustment. Algorithm 1 in Appendix G describes the full online learning procedure.

Figure 6 shows how the topic weights change over time for central bank (CB) retrieval at the one-, three-, six-, and twelve-month horizons. For CB retrieval at the one-month horizon, the weight on demand-related content increases from 0.167 to 0.300. This aligns with previous findings: the share of demand-related text is positively correlated with improvements in forecast accuracy.

In most comparisons, the static correlation between the share of monetary policy content and forecast improvement is negative; however, online learning does not always

[15]We use the squared error rather than the absolute error so that larger prediction deviations are penalized more heavily. The simplex projection follows the algorithm of Wang and Carreira-Perpiñán (2013); see Appendix G for details. The hyperparameters η, κ, and α were set without reference to the evaluation sample. Appendix O examines the sensitivity of final topic weights and retrieved chunks to these parameters at $h = 1$.

lead to a reduction in its weight. In the case of news retrieval, this weight rose from 0.167 to 0.362 for one-month-ahead forecasts, and to 0.437 for three-month-ahead forecasts. For central bank text retrieval, the weights across all forecast horizons remained near or above their initial values.

According to Equation (5), when the reward is positive, the weights shift toward the topic shares $\mathbf{s}_t$, which are weighted by retrieval scores. A positive reward indicates that the squared forecast error falls below the threshold $\kappa = 1.0$. Monetary policy content accounts for a large share of both retrieved central bank texts and news articles. Consequently, when the reward is positive and the weighted share s_t^{mp} exceeds the current weight, the weight assigned to monetary policy increases.

Therefore, the weight of monetary policy content may rise even if such content does not actually contribute to better forecasts. The update rule cannot distinguish which specific topics drive improvements. We suggest testing—using an independent sample—whether reducing or excluding monetary policy-related text might enhance forecast accuracy.

Despite this limitation, dynamic reranking improves forecast performance for both central bank text retrieval and combined retrieval across one-month, three-month, and six-month horizons (see Appendix Table 16). For six-month-ahead forecasts, the RMSE for central bank text retrieval decreased from 0.522 to 0.363, while the RMSE for combined retrieval fell from 0.494 to 0.376.

4.5 Combined Sources: When Mixing Helps vs. Hurts

Panels A through D of Table 4 demonstrate that whether combining central bank texts with news improves forecasts depends on the prompt design and the forecast horizon.

When using "statistical summary" prompts, combining these two types of sources improves one-month-ahead forecasts. The CB+News variant of the statistical text summary template (variant 21) achieves an RMSE of 0.172 at $h = 1$, outperforming both the CB-only variant (0.183) and the news-only variant (0.274).

Combined retrieval also has the lowest RMSE at $h = 1$ in Panel C and at $h = 6$ in Panels B and C. For the twelve-month forecast, all four panels indicate that using news-based retrieval alone yields the lowest RMSE.

When using numerical table prompts, the three-month RMSE for combined retrieval (Variant 14) is higher than that for news-only retrieval (Variant 13)—0.500 versus 0.401 (Table 2).

Therefore, the effectiveness of combining different sources must be evaluated separately for each prompt design and forecast horizon.

Table 3: Initial and Final Topic Weights

		CB-Only ($h = 1$)			News-Only ($h = 3$)		
Channel	Initial	Final	Δ	Static r	Final	Δ	Static r
Demand	0.167	**0.300**	+0.133	0.16	0.206	+0.039	0.20
Energy	0.167	0.218	+0.051	−0.17	0.139	−0.028	**−0.42**
Labor	0.167	0.167	+0.000	−0.04	0.174	+0.007	**0.39**
Housing	0.167	0.009	−0.158	−0.20	0.019	−0.148	−0.12
Supply	0.167	0.016	−0.151	0.09	0.023	−0.144	0.21
Monetary Policy	0.167	**0.290**	+0.123	−0.24	**0.437**	+0.270	**−0.44**
		CB-Only ($h = 6$)			Combined ($h = 6$)		
Channel	Initial	Final	Δ	Static r	Final	Δ	Static r
Demand	0.167	**0.267**	+0.100	0.25	0.249	+0.082	—
Energy	0.167	0.215	+0.048	**0.51**	0.174	+0.007	—
Labor	0.167	0.150	−0.017	0.33	0.173	+0.006	—
Housing	0.167	0.034	−0.133	**−0.36**	0.028	−0.139	—
Supply	0.167	0.021	−0.146	0.34	0.022	−0.145	—
Monetary Policy	0.167	**0.313**	+0.146	**−0.46**	**0.355**	+0.188	—

Notes: All six topic weights start at 1/6 (0.167) and are updated using Equation (5), with learning rate $\eta = 0.25$. The final weights are derived from the last forecast origin that meets the specified criteria. The static correlation coefficient r corresponds to the Pearson correlation coefficient from Table 5, based on the same source and forecast horizon. Values where the final weight exceeds 0.25, or the absolute value of the correlation coefficient exceeds 0.35, are highlighted in bold.
The evaluation period spans from January 2024 to February 2026. Weight updates rely exclusively on actual values observed up to February 2026; consequently, the one-month, three-month, and six-month forecasts underwent 25, 23, and 20 updates, respectively.

5 Comparison with Topic-Model Forecasting

To benchmark the improvements in forecasting achieved in Sections 3.1 through 4 against other text-based prediction methods, we constructed a standard topic modeling-based forecasting pipeline, following the approach of Ellingsen et al. (2022). For this comparison, the RAG-based forecasts use Variants 3 and 4 from Table 2 (based on central bank texts), while the topic modeling approach uses news data.

This method uses Latent Dirichlet Allocation (LDA) to transform a large-scale news corpus into topic-based time series, which are then used as inputs for a machine learning regression model.

We first estimate a Latent Dirichlet Allocation (LDA) model using the GDELT news corpus. The corpus contains 31,018 text chunks covering January 2023 to March 2026. We represent the text using a bag-of-words approach with single words and two-word phrases (unigrams and bigrams). We limit the vocabulary to 10,000 terms and keep only

Table 4: Source-Controlled Comparison: CB vs. News vs. Combined (Same Template, Different Source)

		RMSE by Horizon			
Template Family	Source	$h = 1$	$h = 3$	$h = 6$	$h = 12$
Panel A: Stat summary + RAG summary (variants 19/20/21)					
	CB Only	0.183	**0.392**	**0.270**	0.341
	News Only	0.274	0.403	0.284	**0.281**
	CB + News	**0.172**	0.420	0.307	0.306
Panel B: Stat + RAG summary + table (variants 25/26/27)					
	CB Only	**0.309**	**0.422**	0.303	0.305
	News Only	0.316	0.451	0.301	**0.282**
	CB + News	0.355	0.455	**0.281**	0.283
Panel C: RAG summary + detailed table (variants 28/29/30)					
	CB Only	0.222	0.512	0.323	0.311
	News Only	0.310	**0.421**	0.307	**0.281**
	CB + News	**0.188**	0.478	**0.304**	0.324
Panel D: Stat + RAG + detailed table (variants 31/32/33)					
	CB Only	**0.186**	0.512	0.323	0.311
	News Only	0.310	**0.461**	**0.299**	**0.282**
	CB + News	0.188	0.478	0.305	0.324

Notes: Panels A–D compare text sources under identical prompt formats (variant IDs follow Table 1). Bold indicates the source with the lowest RMSE for each panel and horizon.
Evaluation: January 2024–February 2026. Only outcomes observed by February 2026 are included ($N = 25, 23, 20, 14$ at $h = 1, 3, 6, 12$, respectively).

terms that appear in at least five chunks. The vocabulary and topic model are estimated once on the full corpus and held fixed; only the forecasting regressions are re-estimated at each origin. Our baseline uses $K = 20$ topics, chosen to balance model expressiveness against the constraint that our corpus spans only three years of monthly data: with a lag order of three, the number of predictor variables is $p = 3K$ for text-only forecasts and $p = 3K + 3$ with CPI lags, giving 60 and 63 predictors, respectively. [16]

For each calendar month, we calculate the average topic distribution across text chunks to derive a K-dimensional vector of topic intensities. This monthly topic data serves as a set of predictors for subsequent regression analyses.

[16]We report robustness to $K = 40$ and $K = 80$ in Table 7. The choice of $K = 80$ in Ellingsen et al. (2022) is supported by their 22.5 million articles spanning 35 years. Here, n denotes the number of usable monthly training observations, and p the number of predictor variables, including lagged terms. Increasing the number of topics, K, adds more predictor variables without increasing the number of monthly training observations. Consequently, the ratio of the training sample size to the number of predictor variables, n/p, decreases.

Table 5: Channel–Forecast Improvement Correlations by Source and Horizon

		Correlation with Forecast Improvement			
Source	Channel	$h = 1$	$h = 3$	$h = 6$	$h = 12$
Panel A: Central Bank Documents					
CB	Demand	0.16	**0.47**	0.25	**0.37**
CB	Energy	−0.17	0.19	**0.51**	−0.14
CB	Labor	−0.04	0.22	0.33	**−0.35**
CB	Housing	−0.20	−0.34	**−0.36**	0.31
CB	Supply	0.09	**−0.57**	0.34	0.25
CB	Monetary Policy	−0.24	**−0.41**	**−0.46**	−0.17
Panel B: News Articles					
News	Demand	−0.18	0.20	**0.42**	0.01
News	Energy	0.17	**−0.42**	−0.19	0.27
News	Labor	−0.04	**0.39**	0.01	−0.27
News	Housing	**0.39**	−0.12	0.07	−0.06
News	Supply	0.14	0.21	0.05	−0.13
News	Monetary Policy	−0.34	**−0.44**	**−0.43**	0.30

Notes: Each cell reports the Pearson correlation across forecast origins between a topic's share of retrieved chunks and the forecast gain. Forecast improvement is defined as the absolute error of the baseline model (Variant 1)—which uses only numerical inputs—minus the absolute error of the RAG-based model. A positive correlation indicates that a higher share of a given topic tends to be associated with a greater forecast improvement. Correlation coefficients satisfying $|r| > 0.35$ are shown in bold; this criterion is applied to the values prior to rounding.
The evaluation period spans from January 2024 to February 2026. For forecast horizons $h = 1, 3, 6, 12$, the evaluation includes forecast origins starting from January 2024 and ending in February 2026, December 2025, September 2025, and March 2025, respectively—utilizing only target values observed up to February 2026. The sample sizes for the correlation coefficients of each topic range from 18 to 25, 18 to 23, 17 to 20, and 13 to 14 forecast origins, respectively, depending on the text source and the availability of data for each topic.

At each forecast origin, we estimate two types of regression models using an expanding window of monthly topic proportions:

1. **LASSO (Least Absolute Shrinkage and Selection Operator):** A linear regression model incorporating an L_1 penalty term, with the penalty parameter selected via five-fold cross-validation. This approach follows the regularization method used by Ellingsen et al. (2022).

2. **Random Forest (RF):** A model consisting of 200 trees, with a minimum of two

observations per leaf node. We include this model because Ellingsen et al. (2022) found that Random Forests perform well when using news topics as predictors.

Each model type is implemented using two distinct configurations. The "text-only" specification uses only three lags of each of the K topic proportions; the "text-plus-numerical" specification builds upon this by adding three lagged terms of the year-on-year CPI inflation rate.

The text-only specification corresponds to the text-only ablation experiment for RAG (Variant 3), which provides only text summaries. The text-plus-numerical specification corresponds to the primary RAG setup (Variant 4), which provides both statistical and text summaries. This allows for a comparison of predictive performance with and without historical CPI data. RAG variants 3 and 4 are held fixed across forecast horizons.

We have made three adjustments to the methodology of Ellingsen et al. (2022) to adapt it for monthly inflation forecasting.

First, we forecast year-over-year CPI inflation at monthly frequency rather than quarterly GDP; therefore, there is no need to use Mixed-Data Sampling (MIDAS) regression to integrate data of different frequencies.

Second, we follow their primary specification by using topic frequencies without incorporating sentiment weights. In their robustness analysis, the inclusion of sentiment information worsened consumption forecasts and had little impact on GDP and investment forecasts.

Third, given the smaller scale and shorter time coverage of our news corpus, our baseline specification uses 20 topics instead of 80. We also report results based on 80 topics as a robustness check.

Table 6 compares the predictive accuracy of the LDA-based forecasting models, the RAG variants, and standard econometric benchmarks. To ensure a consistent basis for comparison, all models use the same forecast origins for each horizon. Consequently, Tables 6 and 7 include only those forecast origins for which valid forecasts are available across all models and topic-count specifications. The sample sizes for the one-, three-, six-, and twelve-month forecasts are 21, 19, 14, and 3, respectively.

The RAG forecast values are identical to those in Table 2. The differences in RMSE arise because the comparison with the LDA-based forecasting models uses a smaller set of forecast origins.

Two findings stand out.

When historical CPI data is included, the RAG model achieves a lower RMSE than the LDA benchmarks for the one-, three-, and six-month forecast horizons. This ranking holds true across all three settings for the number of topics (Table 7). For the twelve-month forecast horizon, LDA-RF performs better, though this comparison is limited to only three forecast origins.

Table 6: RAG versus LDA Topic-Model Forecasting: RMSE Comparison

Model	$h = 1$	$h = 3$	$h = 6$	$h = 12$[a]
Panel A: Text only				
LDA-LASSO (news, $K = 20$)	0.413	0.511	0.404	0.239
LDA-RF (news, $K = 20$)	0.285	0.473	0.419	0.201
RAG text-only (variant 3)	2.129	2.051	0.562	1.524
Panel B: Text + numeric				
LDA-LASSO (news + CPI, $K = 20$)	0.258	0.441	0.313	0.442
LDA-RF (news + CPI, $K = 20$)	0.274	0.463	0.387	0.174
RAG stat+text (variant 4)	**0.181**	0.348	**0.264**	0.265
Panel C: Econometric benchmarks				
AR(q)	0.194	**0.323**	0.318	**0.124**
ARIMA	0.202	0.331	0.380	0.183
N (common origins)	21	19	14	3

[a] The twelve-month comparison contains only three forecast origins.

Notes: RMSE is computed over months where realized CPI is available. LDA uses $K = 20$ topics as the baseline specification; Table 7 reports results for $K = 40$ and $K = 80$. Bold entries denote the lowest RMSE across all rows in each column. The RAG variants use central bank text; LDA uses news. AR(q) uses the same AIC-selected AR forecasts as the main comparison.

Evaluation window: January 2024–February 2026. All models in both tables are evaluated on the intersection of valid forecast origins across all LDA specifications at $K = 20, 40, 80$, RAG variants 3 and 4, AR(q), and ARIMA. The common samples contain 21, 19, 14, and 3 observations at $h = 1, 3, 6, 12$, respectively. $h = 1$: origins February 2024–November 2025, targets February 2024–November 2025; $h = 3$: origins April 2024–November 2025, targets June 2024–January 2026; $h = 6$: origins July 2024–September 2025, targets December 2024–February 2026; $h = 12$: origins January 2025–March 2025, targets December 2025–February 2026. Date ranges may contain gaps.

The RAG forecasts are the same as those used in Table 2; their RMSE values differ because this comparison uses a smaller evaluation sample. Variants 3 and 4 are held fixed across horizons.

Retrieved passages may contain economic information not captured by topic proportions, while historical CPI data provides a numerical basis for forecasting. However, this interpretation should be approached with caution, as the comparison involves simultaneous changes to both the text source and the forecasting model.

In the context of text-only forecasting, the results favor LDA. As shown in Panel A, the LDA-based models consistently outperform the RAG text-only configuration (Variant 3, which provides only text summaries) across all forecast horizons. For the one-month forecast, for instance, the RMSE for LDA-RF is 0.285, compared to 2.129 for Variant 3.

This supports the finding from Section 3.1 on the importance of numeric grounding

Table 7: LDA Robustness to Number of Topics

Model	$h = 1$	$h = 3$	$h = 6$	$h = 12$
Panel A: Best LDA model (text + numeric)				
$K = 20$	0.258 ^{L}	0.441 ^{L}	0.313 ^{L}	0.174^{RF}
$K = 40$	0.275 ^{RF}	0.441^{L}	0.337^{L}	0.148 ^{RF}
$K = 80$	0.278 ^{RF}	0.435^{L}	0.313^{L}	0.130 ^{RF}
Panel B: Reference models				
RAG stat+text (variant 4)	**0.181**	0.348	**0.264**	0.265
AR(q)	0.194	**0.323**	0.318	**0.124**
N (common origins)	21	19	14	3

Notes: For each K and horizon, Panel A reports the lower RMSE of LASSO (superscript L) and random forest (superscript RF), using unrounded values. Bold entries denote the lowest RMSE across all rows in each column. The n/p ratio (usable monthly training observations to predictor variables, including lagged terms) ranges from approximately 0.04 with $K = 80$ to 0.48 with $K = 20$ for these text + numeric specifications.
Evaluation window: January 2024–February 2026. The common samples contain 21, 19, 14, and 3 forecast origins at $h = 1, 3, 6, 12$, respectively. Forecast-origin and target-month ranges are reported in Table 6.

for LLM-based forecasting. The LDA pipeline uses observed inflation to estimate the relationship between topic proportions and subsequent inflation.

These results at one, three, and six months are robust to the choice of topic count. Table 7 reports the lower RMSE of LASSO and random forest at each topic count and horizon. At $h = 1$, the best LDA RMSE worsens from 0.258 at $K = 20$ to 0.278 at $K = 80$, consistent with the deteriorating n/p ratio discussed above. At $h = 6$, LASSO achieves an RMSE of 0.313 at both $K = 20$ and $K = 80$. LASSO performs better at three and six months for all three topic counts; random forest performs better at twelve months. At one month, LASSO performs better at $K = 20$, and random forest performs better at $K = 40$ and $K = 80$.

The twelve-month results should be interpreted with caution. They are based on only three common forecast origins, mainly because the news corpus begins in January 2023 and provides a limited training history for longer-horizon regressions. We report these results for completeness, but the sample is too small to draw strong conclusions.

Our LDA benchmark also uses much less data than the original study by Ellingsen et al. (2022), which draws on 22.5 million articles covering 35 years. Differences in the size and length of the training data are important when comparing the performance of the two approaches.

Table 8: Best lead times for text retrieval before the CPI release date

Horizon	CB best cutoff	CB RMSE	News best cutoff	News RMSE
$h = 1$	7 days	0.270	14 days	0.275
$h = 3$	7 days	0.432	14 days	0.400
$h = 6$	30 days	0.477	7–14 days	0.364
$h = 12$	30 days	0.306	0 days	0.308

Notes: Entries show the lowest evaluation-sample RMSE across four lead times (0, 7, 14, and 30 days before the CPI release) for each source and horizon. RMSE is measured in percentage points of year-over-year CPI inflation. Gains are relative to the numeric-only baseline (variant 1 in Table 2), matched by forecast origin and target month, and calculated using unrounded RMSE values.
Evaluation: January 2024–February 2026. Only outcomes observed by February 2026 are included ($N = 25, 23, 20, 14$ at $h = 1, 3, 6, 12$, respectively).

Table 9: Density Forecast Evaluation: CRPS, Coverage, and Interval Width ($h = 1$, Evaluation Sample)

Prompt Variant	CRPS	Log Score	Cov. 68%	Cov. 90%	Width 90%
9: Text + detailed table	0.097	0.352	0.64	0.96	0.694
10: Stat + text + detailed table	**0.097**	**0.355**	0.64	0.96	0.692
7: Stat + text summary + table	0.102	0.322	0.60	0.96	0.714
4: Stat summary + RAG summary	0.112	0.188	0.68	0.96	0.784
2: Stat summary only	0.113	0.192	0.56	0.84	0.662
5: Stat summary + table	0.116	0.137	0.48	0.88	0.622
8: Detailed + table	0.118	0.137	0.48	0.84	0.626
6: Text summary + table	0.128	0.012	0.60	0.88	0.750
12: RAG text extracts (CB)	0.131	−0.333	0.68	0.88	0.788
1: Numeric only	0.154	−0.346	0.56	0.68	0.663
3: Text summary only	0.218	−0.660	0.40	0.80	0.900

Notes: CRPS = Continuous Ranked Probability Score (lower is better). Log Score = mean log predictive density (higher is better). Coverage columns report the fraction of realized values falling within the nominal 68% and 90% prediction intervals. Width 90% is the average width of the 90% prediction interval. Boldface indicates the lowest CRPS and highest Log Score, evaluated before rounding. $N = 25$ forecast origins.
Evaluation window: January 2024–February 2026. Forecast origins and target months both span January 2024–February 2026 at $h = 1$, with the unavailable October 2025 target excluded.

6 Timeliness and Early Signal

6.1 Lead Time Analysis

A practical concern for real-time forecasting is whether text must be contemporaneous with the CPI release to be useful, or whether earlier publications carry durable signal. Table 8 addresses this question by varying the publication date cutoff $\delta \in \{0, 7, 14, 30\}$ days before the CPI release date, restricting retrieved text to documents published at least δ days prior.

Table 10: Narrative disagreement and predictive interval width

Horizon	Entropy Spearman ρ	p-value	Top-decile width	Bottom-decile width
$h = 1$	0.518	0.008	0.580	0.610
$h = 3$	0.597	0.003	0.830	0.760
$h = 6$	0.450	0.046	0.725	0.630
$h = 12$	-0.357	0.211	0.800	0.645

Notes: Combined narrative entropy is the Shannon entropy of the equally weighted average of the central-bank and news topic-share distributions, using natural logarithms. Spearman ρ is the correlation between entropy and the 90% predictive interval width; p-values test zero correlation against a two-sided alternative. Jensen–Shannon (JS) divergence measures the difference between the two source distributions. Top- and bottom-decile widths are mean 90% interval widths at origins with JS divergence at or above the within-horizon 90th percentile and at or below the 10th percentile, respectively. Widths are measured in percentage points of year-over-year CPI inflation.
Evaluation window: January 2024–February 2026. At $h = 1, 3, 6, 12$, evaluated origins start in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively ($N = 25, 23, 20, 14$); only realized targets through February 2026 enter the scores.

Gains are reductions in RMSE relative to the numeric-only forecasts (variant 1) in Table 2, evaluated on common forecast origins and target months and measured in percentage points (pp) of year-over-year CPI inflation.

At $h = 1$, all tested lead times produce negative RMSE gains relative to the numeric-only baseline for both CB and news sources.

We compare four information cutoffs. Table 8 reports the one with the lowest evaluation-sample RMSE for each source and horizon.

The results suggest that the value of recent information differs across sources. At the twelve-month horizon, central bank communications perform best when text is restricted to information available 30 days before the CPI release. Financial news, by contrast, performs best when text is available up to the release date. One possible explanation is that earlier central bank communications contain information that remains useful beyond the most recent release cycle.

6.2 Surprise Months

We also examine whether retrieved text is especially useful when the numeric-only baseline (variant 1) makes large forecast errors. We call these cases "surprise months." At each horizon, we select cases at or above the 90th percentile of the numeric-only baseline's absolute forecast errors.[17] Because there are only a few such months, we use these results to illustrate individual cases rather than draw statistical conclusions.

Table 11 presents selected surprise months at the one-, six-, and twelve-month horizons. For each case, we report the numeric-only forecast and the RAG variant that had

[17]The 90th-percentile thresholds are 0.33 percentage points (pp) at $h = 1$, 0.55 pp at $h = 3$, 0.84 pp at $h = 6$, and 0.58 pp at $h = 12$. This identifies 2–3 surprise months per horizon.

the smallest absolute error in hindsight.

One example is the April 2024 forecast origin at the twelve-month horizon. The numeric-only model predicts inflation of 3.65% for March 2025, while actual inflation is 2.38%, giving an error of 1.27 percentage points. The central-bank-only RAG variant (variant 12) predicts 2.43%, with an error of just 0.05 percentage points. In this case, the text-augmented forecast is much closer to the realized inflation rate and better reflects the disinflation that followed.

The selected RAG forecasts have smaller errors than the baseline in all seven cases shown. At $h = 6$, both surprise origins involve the baseline forecasting 3.44 percent while actuals came in around 2.4–2.6 percent; the RAG variant selected *ex post* at each origin reduces the error from roughly 1 pp to approximately 0.02–0.04 pp. At $h = 1$, the selected RAG forecasts reduce the baseline error by more than half in each of the three cases. The largest absolute error reductions in the table occur at six and twelve months.

In contrast to these *ex post* selected cases, Section 3.1 evaluates the performance of each fixed-configuration forecasting approach across the full sample.

7 Density Forecasting and Uncertainty

7.1 Density Forecast Calibration

The accuracy of point forecasts reflects only one dimension of forecast quality. For purposes such as risk management and policy analysis, the full predictive distribution is equally important. We ask the LLM to report the 5th, 16th, 50th, 84th, and 95th percentiles, and we evaluate the density forecasts using the Continuous Ranked Probability Score (CRPS) and interval coverage.

Table 9 reports the results for one-month-ahead forecasts. For a prediction interval with a nominal coverage of 90%, the actual coverage is 68% for the raw numeric-only prompt (Variant 1) and 84% for the detailed numeric-only prompt (Variant 8). Variants 9 and 10, which incorporate text, achieve a coverage of 96%. Figure 7 compares coverage rates across different horizons, while Appendix E provides the CRPS and coverage results for longer horizons.

CRPS also showed improvement. The raw numeric-only prompt yielded a CRPS of 0.154, while variants 9 and 10 reduced this to 0.097—a decrease of approximately 37%. The detailed numeric-only prompt (variant 8) resulted in a CRPS of 0.118. These comparisons indicate that both the presentation of numerical data and the retrieved text influence the performance of density forecasts.

The text-only prompt (variant 3) serves as an interesting point of comparison. It achieved an actual coverage rate of 80% for the 90% prediction interval—higher than the 68% observed with the raw numeric-only prompt—but also yielded a higher CRPS

Table 11: Surprise Months: Numeric-Only Baseline vs. Best RAG Variant

Horizon	Origin	Target	Actual	Baseline (v1)	Best RAG
$h = 1$	2024-01	2024-01	3.09	3.44	2.93 ^{v4}
$h = 1$	2025-01	2025-01	2.99	2.53	2.93 ^{v4}
$h = 1$	2025-03	2025-03	2.38	2.84	2.45 ^{v3}
$h = 6$	2024-04	2024-09	2.43	3.44	2.45 ^{v3}
$h = 6$	2024-05	2024-10	2.58	3.44	2.54 ^{v7}
$h = 12$	2024-04	2025-03	2.38	3.65	2.43 ^{v12}
$h = 12$	2024-10	2025-09	3.02	2.44	2.83 ^{v4}

Notes: Surprise months are origins where the numeric-only baseline (variant 1) absolute error falls at or above the 90th percentile at the given horizon. All values are year-over-year CPI inflation in percentage points. The superscripts follow the variant numbering used in the paper: $v3$ denotes the use of text summaries only; $v4$ denotes statistical summaries combined with text summaries; $v7$ denotes statistical summaries, text summaries, and numerical tables; and $v12$ denotes central bank text summaries combined with numerical tables.

For each combination of forecast origin and forecast horizon presented in the table, we select the RAG (Retrieval-Augmented Generation) forecast—from variants 3, 4, 6, 7, 9, 10, and 12—that yields the lowest actual absolute error. This selection is made *ex post*, following the observation of actual outcomes; in the event of a tie in error magnitude, the variant with the lowest index number is chosen.

The evaluation period spans from January 2024 to February 2026. The seven cases presented here feature forecast origins ranging from January 2024 to March 2025, with target months ranging from January 2024 to September 2025. This set includes three cases for one-month-ahead forecasts, and two cases each for six-month-ahead and twelve-month-ahead forecasts.

(0.218 versus 0.154). Thus, an increase in coverage does not necessarily imply better overall density forecasting.

7.2 Source Disagreement and Forecast Uncertainty

We examine whether the topics present in the retrieved text are associated with forecast uncertainty. Topic entropy measures the uniformity of the topic distribution within the text, while the Jensen–Shannon (JS) divergence quantifies the difference in topic composition between central bank texts and news sources. Table 10 reports the relationship between these metrics and the width of the 90% prediction intervals.

For one-, three-, and six-month forecasts, broader topic coverage is generally associated with wider prediction intervals. When the text encompasses a variety of economic pressures, the LLM may account for a wider range of potential outcomes. For three-month forecasts, instances where there is a greater divergence in topics between the two sources also correspond to wider prediction intervals. Here, "divergence" refers to differences in topic coverage; it does not necessarily imply that the two sources hold conflicting views on inflation.

For twelve-month forecasts, the correlation between topic entropy and interval width becomes negative, though it lacks statistical significance. Evidence for this time horizon is, therefore, less conclusive.

7.3 Relation to Quantile Regression Forest Approaches

Lenza et al. (2025) use Quantile Regression Forests (QRF) to forecast the probability distribution of inflation in the Eurozone, achieving performance competitive with both linear and non-linear benchmark models.

This paper likewise investigates density forecasting, but focuses on U.S. CPI data. We examine how the presentation of numerical data and the content of retrieved text influence the LLM's predictive distribution. Our results indicate that both factors affect interval coverage and the Continuous Ranked Probability Score (CRPS).

A potential extension involves combining Quantile Regression Forests (QRF) with retrieved text to investigate whether such text can provide additional predictive value for density forecasting beyond what is already captured by numerical indicators.

8 Conclusion

We investigate whether economic texts can improve inflation forecasts and how their predictive value varies by source, topic, and forecast horizon. Using Retrieval-Augmented Generation (RAG), we provide an LLM with central bank communications, financial

news, and historical U.S. CPI data. The best-performing RAG configurations—measured by either RMSE or MAE—primarily enhance forecast accuracy at six-month and twelve-month horizons. For six-month forecasts, improvements are observed across most time periods, whereas for twelve-month forecasts, the largest gains are concentrated in a few specific periods. Results for shorter horizons differ: the ranking of one-month forecasts depends on the chosen evaluation metric, while for three-month forecasts, an AR(q) model performs better. When prompts combine statistical summaries with retrieved text, the improvements over selected univariate benchmarks are statistically significant at the 5% level for both six-month and twelve-month horizons.

Textual data can complement historical inflation data, though its value depends on the source and the forecast horizon. In our sample, central bank communications perform better for one-month forecasts, while financial news is more effective for twelve-month forecasts. However, when using prompts that incorporate statistical summaries, the performance difference between these two sources is not statistically significant at the 5% level. For twelve-month forecasts, improvements persist even when documents published within 30 days prior to the forecast origin are excluded from the retrieval process. The presentation of numerical data also influences both point and density forecasts; while some improvements in density forecasting stem from these adjustments, the actual coverage of twelve-month prediction intervals remains below the nominal level. Across most sources and horizons, a higher share of monetary policy-related text tends to be associated with smaller improvements in forecast accuracy.

While these experiments do not reveal the internal reasoning processes of the model, they demonstrate how forecasts shift in response to variations in numerical inputs and retrieved text. By keeping the prompt format constant, we can compare the impact of different text sources; by altering the presentation of numerical data, we can observe how the model responds to raw tabular data versus statistical summaries. Topic analysis provides evidence regarding the relationship between the retrieved content and improvements in forecasting.

Forecasters can adjust numerical summaries and text sources without the need to retrain the model. Compared to raw tables, concise statistical summaries have the potential to enhance forecast accuracy. When selecting text sources, the forecast horizon should also be taken into account.

Our results are based on a single model, Llama-3-8B-Instruct; outcomes might differ if other models were used. While the model's weights are publicly available and the pre-training cutoff date is documented, the cutoff date for post-training data has not been disclosed. Appendix K examines the model's knowledge of historical CPI releases. Although these tests help gauge what information the model possesses, they cannot determine the full scope of its training data.

The primary forecast comparisons include only 14 to 25 forecast origins per horizon.

This limits the precision of the comparisons and makes average performance sensitive to individual forecast errors. Furthermore, the analysis is restricted to U.S. CPI inflation; therefore, the ranking of text sources—and how these rankings shift across different horizons—may not necessarily apply to other economies. Moreover, topic correlations do not constitute causal evidence; one cannot conclude that excluding texts related to monetary policy would necessarily improve forecasts.

Future work could involve comparing different topic-filtering methods on independent samples to test whether reducing or removing texts concerning monetary policy improves forecast accuracy. Future research could also investigate whether these findings hold for larger language models. Applying this framework to other economies would require tailoring forecast targets, data sources, and language-processing methods to local contexts (see Appendix Q). Extending the evaluation period would also facilitate more precise comparisons across different sources and horizons, while allowing for an assessment of whether any observed improvements in forecasting are sustained over time.

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Figures

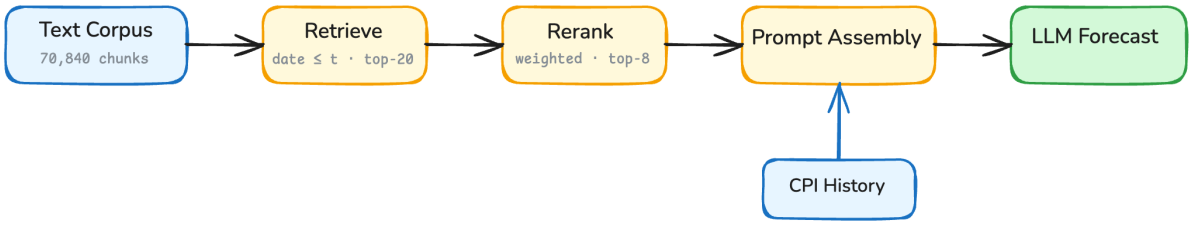


Figure 1: RAG pipeline for a single forecast origin. Blue boxes denote data inputs; yellow boxes denote processing stages; the green box is the final forecast output. Retrieval excludes chunks from documents published after forecast origin t.

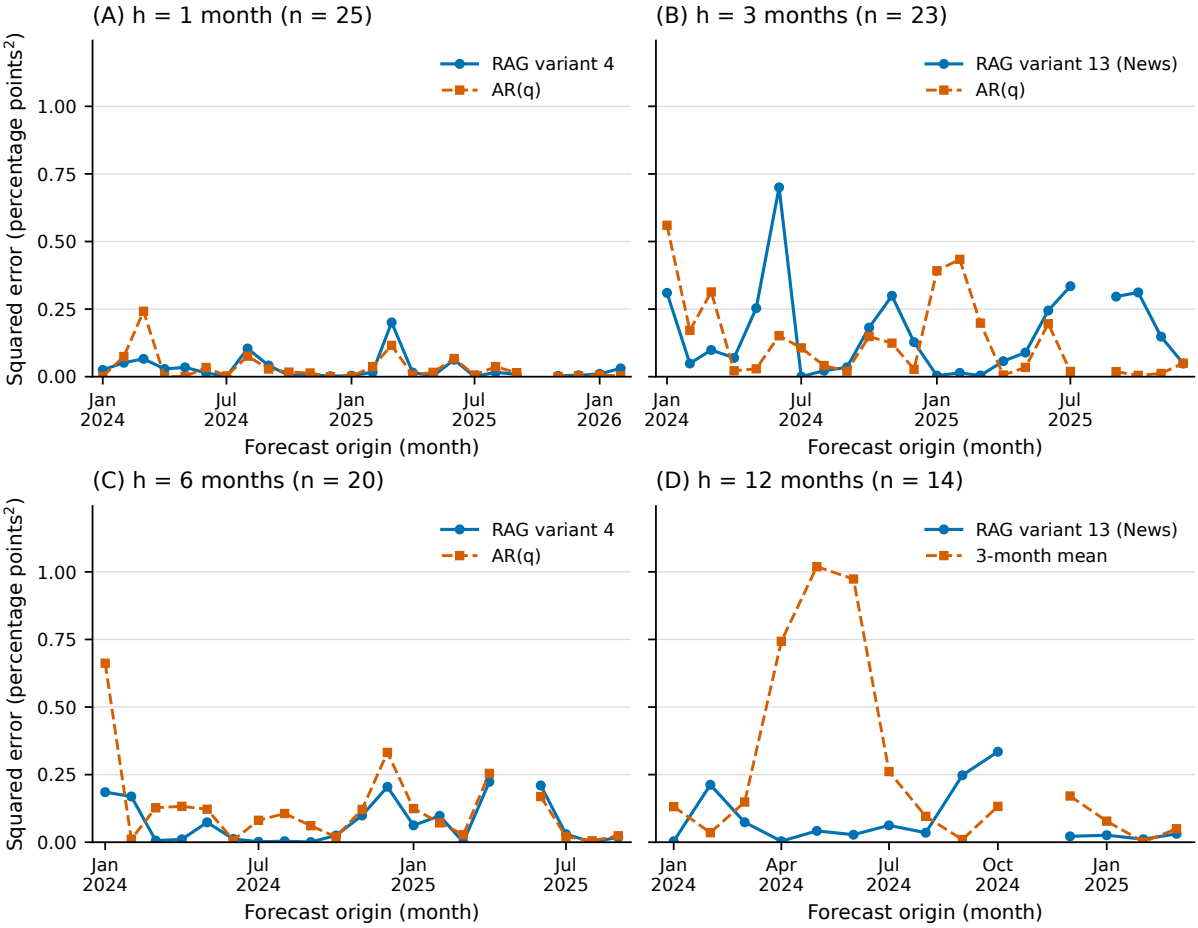


Figure 2: Squared Forecast Errors Across Forecast Horizons. Panels A–D plot unsmoothed squared forecast errors at $h = 1, 3, 6, 12$ on a common scale, in squared percentage points. RAG candidates are the text-using variants 3, 4, 7, 9, and 11–14 in Table 2. Univariate candidates are AR(q), ARIMA, last-observed, three-month mean, seasonal naïve, and Atkeson–Ohanian forecasts where available. Within each model family and horizon, we select the model with the lowest evaluation-sample RMSE and hold it fixed across forecast origins. The selected models are RAG variant 4 at $h = 1, 6$ and variant 13 at $h = 3, 12$; AR(q) at $h = 1, 3, 6$ and the three-month mean at $h = 12$. All panels begin in January 2024; the final origins are February 2026, December 2025, September 2025, and March 2025, respectively, yielding 25, 23, 20, and 14 paired observations. Pairs share the origin, target month, and actual value; gaps indicate unavailable targets. Target months span January, March, June, and December 2024, respectively, through February 2026.

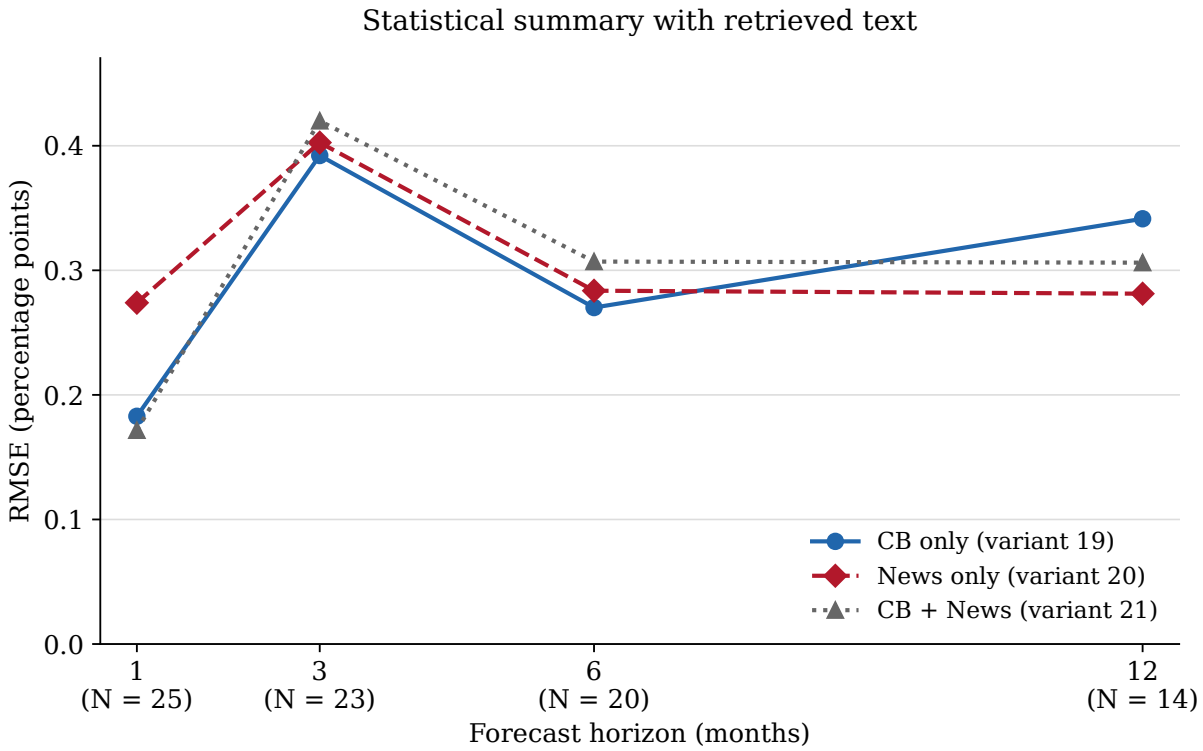


Figure 3: Forecast Accuracy by Retrieval Source. The figure reports RMSE, measured in percentage points of year-over-year CPI inflation, for the statistical-summary specification with CB-only, news-only, and combined retrieval (variants 19, 20, and 21). The numerical inputs and prompt structure are held fixed across sources. Results correspond to Panel A of Table 4. The January 2024–February 2026 evaluation uses 25, 23, 20, and 14 common forecast origins at $h = 1, 3, 6, 12$, respectively. Origins begin in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively; target months begin in January, March, June, and December 2024, respectively, and end in February 2026. Only observed targets enter the scores.

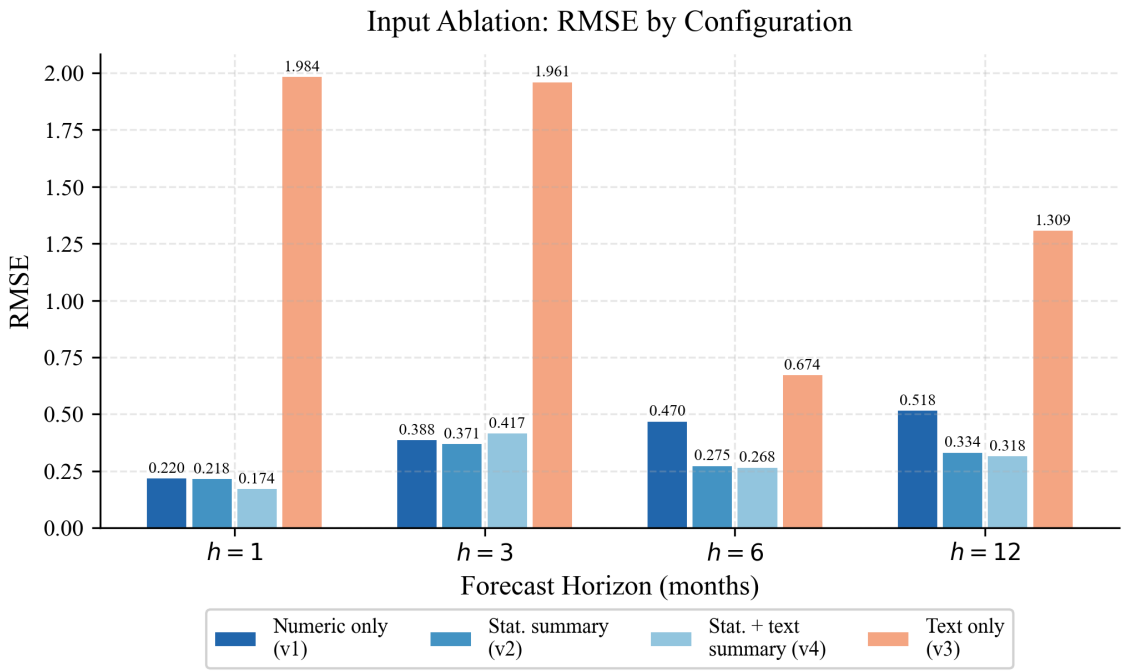


Figure 4: Input Ablation: RMSE by Configuration. For each forecast horizon, a grouped bar chart compares four prompt configurations: numerical tables only (Variant 1), statistical summaries (Variant 2), statistical summaries combined with text summaries (Variant 4), and text only (Variant 3). For the one-month, six-month, and twelve-month forecasts, Variant 4 yields a lower RMSE than the baseline numerical table configuration (Variant 1), with the largest improvements observed at the six-month and twelve-month horizons. The results for the three-month forecast are the opposite, with Variant 4 showing a higher RMSE. This comparison involves variations in both the presentation of numerical data and the provided text input. Across all four forecast horizons, the text-only configuration results in a higher RMSE than configurations that include numerical input. The values atop the bars represent the RMSE, measured in percentage points. Evaluation window: January 2024–February 2026. At $h = 1, 3, 6, 12$, evaluated forecast origins start in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively ($N = 25, 23, 20, 14$). Only target outcomes observed by February 2026 enter the evaluation.

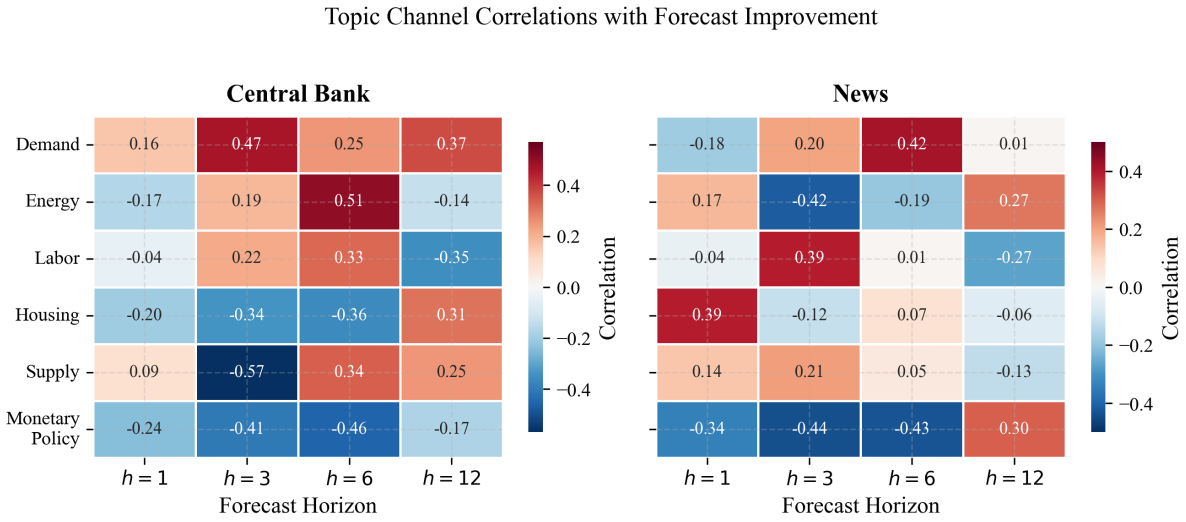


Figure 5: Topic Channel Correlations with Forecast Improvement. Each cell reports the Pearson correlation across forecast origins between a topic's share of retrieved chunks and forecast improvement, for a given source and horizon. Forecast improvement is defined as the absolute error of a baseline model (Variant 1)—which uses only numerical inputs—minus the absolute error of the RAG-based model. The prompts combine CPI numerical tables with either central bank-sourced text (Variant 12) or news-sourced text (Variant 13); the correlation coefficients correspond to those presented in Table 5. The left panel displays results for central bank documents, while the right panel covers news articles. For both types of sources, the share of "demand" topics is positively correlated with improvements in three-month and six-month forecasts. Across most combinations of sources and time horizons, the share of "monetary policy" topics is negatively correlated with forecast improvement, though a positive correlation is observed for twelve-month forecasts based on news sources. The evaluation period spans from January 2024 to February 2026. For forecast horizons $h = 1, 3, 6, 12$, eligible forecast origins begin in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively—using only the target values observed up to February 2026. The sample sizes for the correlation coefficients of each topic range from 18 to 25, 18 to 23, 17 to 20, and 13 to 14 forecast origins, respectively, depending on the text source and the availability of data for each topic.

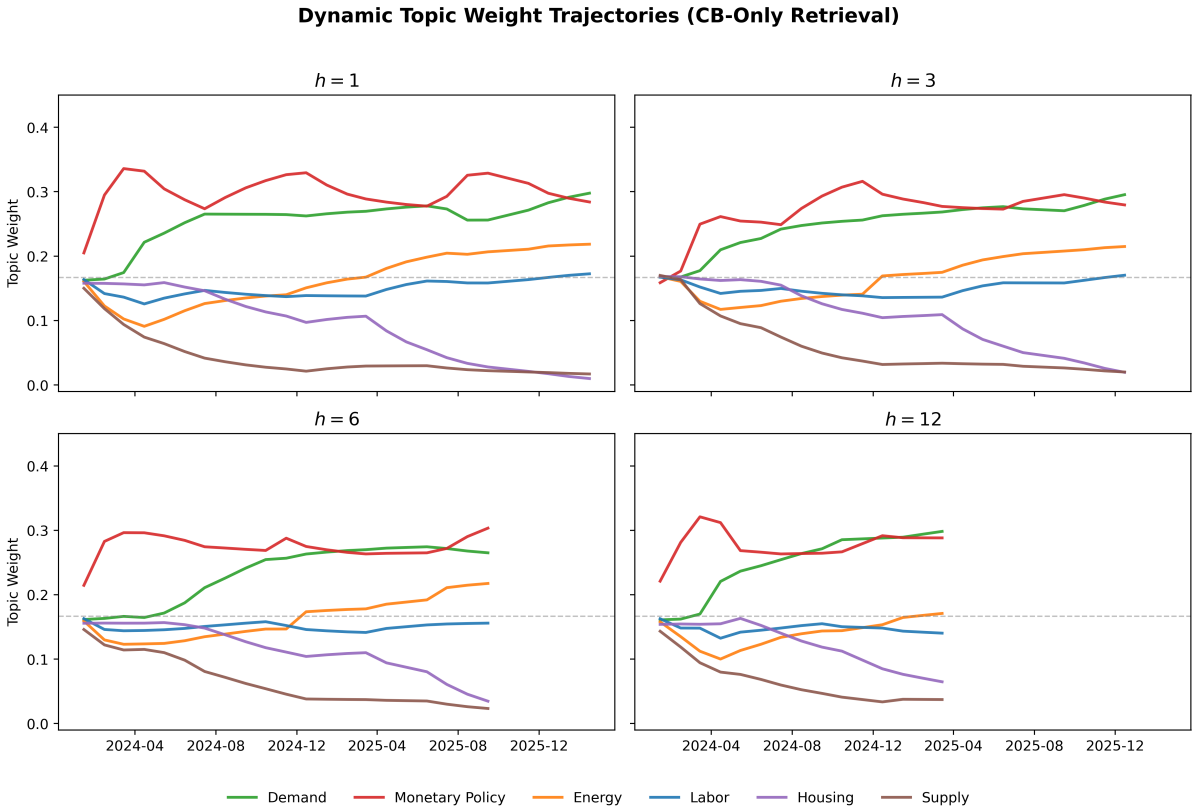


Figure 6: Dynamic Topic Weight Trajectories Over the Evaluation Window. Each curve represents the evolution of the weight assigned to a specific topic. The online learner updates these weights following the release of the actual outcomes for each forecast. The initial weight for every topic is $1/6 \approx 0.167$, which is indicated in the plots by a horizontal dashed line. All four panels are based on text retrieved from central bank sources and correspond to one-month, three-month, six-month, and twelve-month-ahead forecasts, respectively. Across all horizons, the final weights for "demand" and "monetary policy" exceed their initial values, while the weights for "housing" and "supply" fall below them. Notably, the weight for "monetary policy" increases despite a negative static correlation, as shown in Table 5. Evaluation window: January 2024–February 2026. At $h = 1, 3, 6, 12$, evaluated forecast origins start in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively ($N = 25, 23, 20, 14$). Only target outcomes observed by February 2026 enter the weight updates.

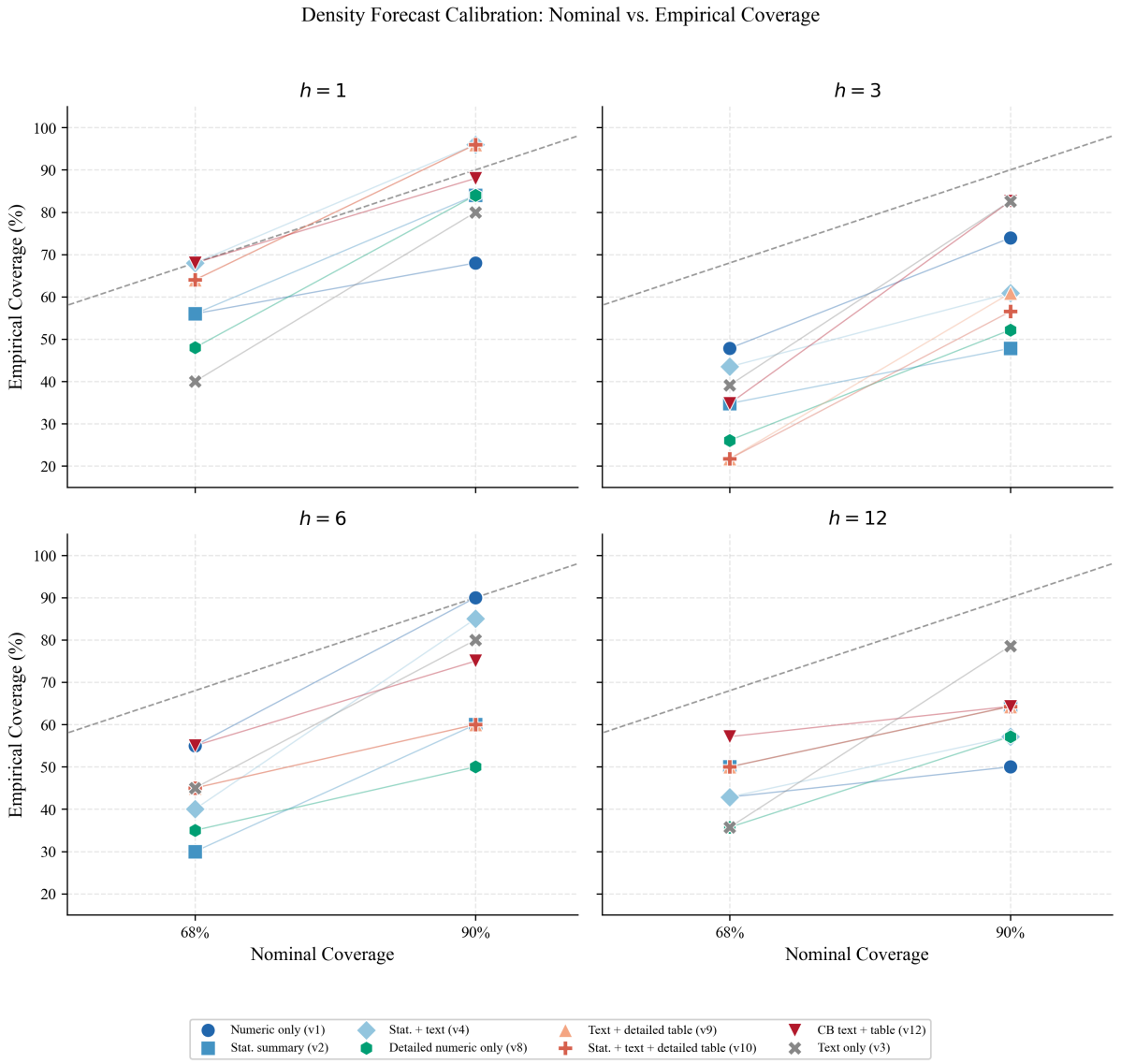


Figure 7: Density Forecast Calibration: Nominal vs. Empirical Coverage. Each panel corresponds to a specific forecast horizon ($h = 1, 3, 6, 12$). For each prompt variant, two points—connected by a line segment—represent the empirical coverage rates for the 68% and 90% prediction intervals. Points falling on the 45-degree dashed line indicate that the empirical coverage matches the nominal level, while points below the line indicate undercoverage. For one-month-ahead forecasts, variants 4, 9, and 10—which incorporate textual information—achieved an empirical coverage rate of 96% for intervals with a nominal coverage level of 90%. In contrast, variant 8 (using only detailed numerical information) yielded 84%, and variant 1 (using only raw numerical tables) yielded 68%. For twelve-month-ahead forecasts, the empirical coverage rates for all configurations shown in the figure fall below both nominal levels. The text-only configuration (variant 3) exhibits undercoverage for both types of intervals across all four forecast horizons, with a more pronounced shortfall observed for the 68% prediction intervals. Evaluation window: January 2024–February 2026. At $h = 1, 3, 6, 12$, evaluated forecast origins start in January 2024 and end in February 2026, December 2025, September 2025, and March 2025, respectively ($N = 25, 23, 20, 14$). Only target outcomes observed by February 2026 enter the evaluation.